

PREDICTION ON CALORIES BURNED BASED ON JOGGING ACTIVITY

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Abstract— The rise of sedentary lifestyles has heightened the demand for advanced fitness tools that provide precise and actionable insights into calorie expenditure. This project introduces a machine learning-based application designed to predict energy expenditure by leveraging user-specific data, including age, weight, height, and activity type. The system utilizes metabolic equivalent of task (MET) values and a continuously learning database to ensure accurate and personalized predictions. By integrating goal tracking and tailored recommendations, the platform supports users in achieving health objectives such as weight loss, muscle gain, or maintenance. This scalable and user-friendly solution addresses the limitations of traditional calorie-tracking methods, empowering individuals to make informed decisions and fostering long-term adherence to personalized fitness goals.

Keywords— Calories Burn Prediction, Machine Learning, Fitness Tool, Personalized Health, Calorie Tracking, XGBoost, Fitness Goals

I. EXECUTIVE SUMMARY

Calories Burn Prediction addresses the growing need for accurate, personalized fitness tools in response to rising sedentary lifestyles. Leveraging machine learning, the application provides precise calorie expenditure predictions based on user-specific attributes such as age, weight, height, gender, and activity type. This tailored approach empowers users to achieve health goals like weight loss, muscle gain, or maintenance through actionable insights and intuitive goal tracking.

Central to the platform is a dynamic, learning-driven database that refines predictions over time. By integrating

metabolic equivalent of task (MET) values, the system ensures accuracy across diverse activities. Its user-friendly design simplifies navigation, while its scalable architecture enables continuous improvement and adaptation to emerging health trends.

This project bridges the gap between intention and action in fitness, equipping individuals with a powerful tool to make informed, sustainable health decisions.

II. PROJECT BACKGROUND

As sedentary lifestyles rise, the need for effective fitness tools becomes critical. Traditional calorie-tracking methods often use generic estimates that fail to reflect individual differences. This project was developed to fill this gap with a machine learning-based solution that provides precise, personalized insights.

By combining user-specific information with a database of metabolic data, Our project delivers tailored recommendations for achieving health goals. Its focus on continuous learning and personalization ensures its relevance to diverse user needs. Calories Burn Prediction addresses the demand for accurate and actionable fitness

solutions, empowering individuals to take control of their wellness journey.

III. OBJECTIVE

The primary objective of this project is to develop a user-centric application designed to accurately predict calorie expenditure by leveraging individual attributes—such as age, weight, height, and fitness level—along with specific activity types. By integrating personalized data and advanced algorithms, the app aims to offer users precise insights into their energy expenditure, enabling them to make informed decisions about their fitness and weight management routines. Beyond predictions, the application will provide actionable recommendations tailored to each user's goals, whether they aim to lose weight, gain muscle, or maintain a healthy lifestyle. To ensure continuous improvement and relevance, the project also focuses on building a comprehensive database of human activity data, which will serve as a foundation for refining predictive models and enhancing user experiences over time. Ultimately, the app seeks to empower individuals with an intuitive, accessible tool that simplifies goal-setting, tracks progress, and fosters long-term adherence to personalized health objectives, thereby bridging the gap between intention and achievement in wellness journeys.

IV. PROJECT SCOPE & SIGNIFICANT

The project scope encompasses the development of a fitness application designed to collect and process user-specific data—including age, gender, height, weight,

activity type, and fitness goals—to deliver personalized health insights. Central to its functionality is a machine learning-driven calorie prediction system that calculates energy expenditure with precision by analyzing metabolic equivalent of task (MET) values across diverse activities. The app will further incorporate goal-tracking features, allowing users to set, monitor, and adjust fitness objectives while receiving tailored recommendations to stay on course. A dynamic, anonymized database will underpin the platform, continuously refining prediction accuracy as it aggregates activity data over time. To ensure accessibility, the application will prioritize user-centric design through an intuitive interface that simplifies navigation and interaction.

The significance of this initiative lies in its response to the growing demand for data-driven fitness solutions. By prioritizing personalization, the app adapts recommendations to individual needs, fostering greater engagement and adherence. Machine learning integration enhances the accuracy of calorie burn estimates, addressing a common limitation in generic fitness tools. Beyond metrics, the platform empowers users to take ownership of their health through actionable insights and progress tracking, bridging the gap between intention and action. Additionally, its scalable architecture ensures the system evolves alongside expanding datasets, improving predictions and adaptability to emerging health trends. Collectively, these elements position the application as a transformative tool for promoting informed, sustainable health decisions.

V. PROPOSED SOLUTION

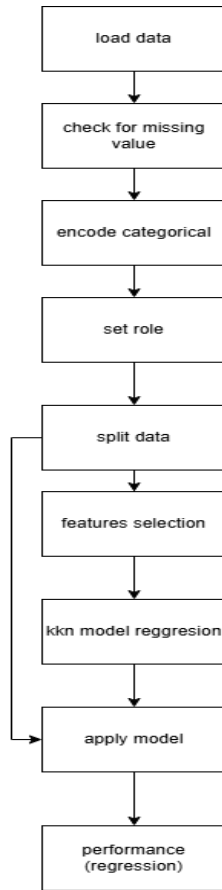


Figure 1.1 illustrate for model flow

This project implements a robust calorie expenditure prediction system using an XGBoost Regressor. The workflow begins with data ingestion from a Calorie.csv dataset, followed by data preparation involving categorical encoding and feature scaling. To identify the most relevant predictors, a hybrid feature selection approach is used, combining Recursive Feature Elimination (RFE) with

Linear Regression and Lasso Regression, based on top features. The core of the model development involves a hyperparameter-tuned XGBoost Regressor, optimized via a GridSearchCV search across a defined space of parameters including `n_estimators`, `max_depth`, `learning_rate`, `subsample`, and `colsample_bytree`. This model, selected based on the best cross validation score, is then trained on a training data set and assessed using metrics including MAE, MSE, and R2. Finally, feature importance is visualized using `plot_importance` to provide insights into the model's decision-making process. This approach demonstrates a full cycle of model development, including selection, tuning, evaluation and model interpretability.

The jogging dataset collected at UTeM offers a unique opportunity to integrate physical activity metrics into predictive health models. Regular jogging is not only a widely recognized form of exercise but also an important factor to maintaining cardiovascular health. By incorporating this dataset into machine learning frameworks, several benefits can be realized.

Firstly, jogging-related variables such as frequency, distance, time, and calorie burn can significantly enhance the predictive accuracy of heart disease risk models. These metrics provide a direct measure of physical activity, which is a key determinant of overall cardiovascular health. For example, consistent jogging routines that improve BMI and calorie burn

can be used as indicators of lower risk profiles, providing valuable input for risk assessment.

Secondly, the dataset highlights the preventive benefits of physical activity. By analyzing how regular jogging impacts BMI and calorie expenditure, the model can showcase actionable insights into lifestyle changes that reduce cardiovascular risks. This aligns with the broader goal of promoting preventive healthcare strategies and raising awareness of the health benefits of consistent exercise.

Third, XGBoost is a scalable, ensemble-based gradient-boosting framework designed for efficiency and high performance in both regression and classification tasks. Unlike KNN, which relies on local averaging, XGBoost builds sequential decision trees where each tree corrects the errors of its predecessor. It minimizes a loss function (e.g., mean squared error for regression) using gradient descent and incorporates regularization terms (λ and α) to prevent overfitting. Key features include handling missing data, parallel processing, and tree-specific parameters like `max_depth` (tree complexity) and `learning_rate` (step size for corrections). While not used in the provided code, XGBoost is widely favored for structured datasets due to its speed and accuracy, often outperforming simpler models like KNN. For example, in calorie prediction, XGBoost could capture complex interactions between features like `bmi`, `time`, and `distance` through its ensemble of

trees. Its flexibility and robustness make it a popular choice in machine learning competitions and industrial applications.

Lastly, this dataset paves the way for more personalized healthcare interventions. By understanding individual performance metrics, such as jogging speed and stamina, tailored exercise plans can be developed to optimize health outcomes. Personalized recommendations not only improve patient adherence but also foster proactive engagement with fitness routines.

In conclusion, the jogging dataset enhances the project's ability to address cardiovascular health from a preventive and personalized perspective. Its integration into predictive models bridges the gap between data analytics and real-world health interventions, setting a strong foundation for comprehensive and actionable healthcare solutions. Additional Features

While the dataset is localized to UTaM and focuses on evening jogging sessions, the inclusion of calculated variables like BMI and calorie expenditure enriches the analysis. However, the absence of data from other demographics, such as female participants or individuals from other locations, limits the generalizability of the model.

VI. MACHINE LEARNING EXPERIMENT

In this study, six regression models were tested to evaluate their predictive performance on a given dataset. The

models included Multiple Linear Regression (MLR), Support Vector Regression (SVR), Gradient Boosting Regressor (GBR), Random Forest Regressor (RFR), Voting Regressor, and Stacking Regressor. MLR was implemented using its default configuration, assuming a linear relationship between the input features and target variable. SVR was applied with a radial basis function (RBF) kernel, using default values for regularization and epsilon parameters. GBR and RFR were tested with default configurations, without modifications to the number of estimators, learning rate, or tree depth. The Voting Regressor aggregated the predictions of MLR, SVR, GBR, and RFR by simple averaging, while the Stacking Regressor combined the predictions from these base models and trained a meta-model (linear regression) to improve overall performance.

The models were evaluated using two primary metrics: Mean Absolute Error (MAE) and R-squared (R^2). MAE calculates the average squared difference between predicted and actual values, where a lower value indicates better predictive performance. R^2 represents the proportion of variance explained by the model, with higher values indicating better model fit. To assess generalization ability and avoid overfitting, 5-fold cross-validation was employed. In this process, the dataset was divided into five subsets, with each subset used once for testing while the remaining four were used for training. This procedure was repeated five times to ensure the reliability of the

results. Additionally, hyperparameter tuning was performed for models such as SVR, GBR, and RFR using grid search to optimize parameters, including the number of estimators, learning rate, and regularization factors.

The methodology aimed to assess the performance of both individual models and ensemble methods. Hyperparameter tuning and cross-validation were employed to optimize model configurations and ensure robust and reliable results. This experimental setup provided a comprehensive comparison of the models' performance in terms of predictive accuracy and generalization ability.

VII. RESULT DISCUSSION

Mean Absolute Error (MAE) is a commonly used metric in machine learning to measure the average magnitude of errors in a set of predictions, without considering their direction. It is calculated as the mean of the absolute differences between predicted values and actual values. MAE provides a straightforward interpretation by expressing the average error in the same units as the target variable, making it particularly useful for understanding how far off predictions are on average. A lower MAE indicates better predictive accuracy.

R-squared (R^2) is a statistical metric used in machine learning and

regression analysis to evaluate the proportion of the variance in the target variable that is explained by the predictive model. It ranges from 0 to 1, where the value of 1 indicates that the model explains all the variance in the target variable and the value of 0 indicates that the model explains none of the variance. While the R^2 value of negative means that the model performs worse than a horizontal line at the target variable. R^2 is a measure of how well the regression model fits the data and is often used to assess the model's goodness of fit.

In evaluating the K-Nearest Neighbors (KNN) Regressor Model, the performance metrics reveal an MAE of 68.40 and an R^2 value of 0.26. The relatively low R^2 indicates that the model explains only 26% of the variance in the target variable, highlighting its limited ability to capture underlying patterns in the data. Despite this, the lower MAE suggests that the model achieves smaller average prediction errors, providing consistency in its predictions. However, its overall predictive power remains constrained by the simplicity of the KNN algorithm.

Conversely, the XGBoost Regressor Model exhibits a higher R^2 value of 0.69, signifying that it explains a greater proportion of the variance in the target variable compared to KNN. This highlights XGBoost's ability to model complex relationships in the data. However, the model records a higher MAE of 92.74, indicating that its

predictions, while more aligned with the overall variance, suffer from larger individual errors. This discrepancy suggests a trade-off between capturing general patterns and minimizing average prediction errors.

The K-Nearest Neighbors (KNN) Regressor faces several challenges that may limit its effectiveness when applied to the Calorie dataset. First, KNN relies heavily on local patterns and may struggle to generalize complex, global relationships in the data. Additionally, the algorithm is sensitive to feature scaling, meaning differences in scale among features can skew distance calculations and reduce predictive accuracy. Finally, KNN does not inherently capture feature interactions, limiting its capacity to model the nuanced relationships between variables in the dataset. Addressing these issues through preprocessing, hyperparameter tuning, and dimensionality reduction could improve the model's performance.

In conclusion, the XGBoost Regressor emerges as the superior model in terms of explanatory power (R^2), while the KNN Regressor offers better consistency (lower MAE). Future enhancements in feature selection, hyperparameter tuning, and data preprocessing could further improve the performance of both models.

VIII. CONCLUSION

The Calories Burn Prediction project demonstrates the potential of machine learning to revolutionize fitness management

by offering accurate, personalized insights into calorie expenditure. By integrating user-specific data with metabolic equivalent of task (MET) values, the application provides tailored recommendations to help individuals achieve their health goals, whether they aim to lose weight, gain muscle, or maintain a balanced lifestyle.

The evaluation of various machine learning models, including XGBoost, highlights the importance of advanced algorithms in capturing complex patterns and delivering reliable predictions. The dynamic database and user-friendly interface ensure that the platform remains scalable, adaptive, and relevant to diverse user needs over time.

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APPENDIX

#	A	B	C	D	E	F	G	H	I
1	Gender	age	weight	height	frequencies	distance	calories	bmi	time
2	male	21	55	164	3	5	260	20.449137	0.5
3	male	24	53	163	4	7	150	19.94806	1
4	male	22	55	164	3	4.8	348	20.449137	1
5	male	20	48	160	4	4	300	18.75	0.5
6	male	22	74	168	3	5	386	26.218821	0.5
7	male	22	70	176	3	5	450	22.598141	0.5
8	Female	22	40	155	3	6	269	16.649324	0.85
9	male	20	69	162	3	3	300	26.291724	0.75
10	male	23	74	168	4	3	150	26.218821	0.5
11	male	22	65	173	3	3	346	21.718066	0.5
12	male	23	72	173	1	2	294	24.056935	0.5
13	male	20	54	161	4	4	227	20.83253	0.5
14	male	20	59.72	173.27	3	1.5	95	19.891753	0.25
15	male	19	60	169	2	11	433	21.807668	1.5
16	Female	21	53	158	2	5	250	21.30572	1.5
17	Female	19	53	157	1	6	321	21.501886	1
18	male	21	65	176	1	5	525	20.983988	0.5
19	male	19	58	163	2	5	277	21.829952	1
20	Female	22	42	150	3	3	111	18.666667	1
21	male	18	55	170	3	3	197	19.031142	0.33
22	male	22	60	170	4	7.5	323	20.761246	0.5
23	Female	20	39	149	1	1.5	139	17.566776	0.42
24	male	20	55	173	4	2	50	18.376825	0.33
25	male	19	59	170	3	5	191	20.415225	0.5
26	male	22	70	170	3	2.4	250	24.221453	0.5
27	male	23	90	180	2	4	360	27.777778	0.5

Figure 1.2 table of dataset for jogging

PROJECT ML: JOGGING COUNTS
CALORIES P1

Hello, we are a group working on a machine learning project to create calories counter. We would like to gather data from joggers, and we kindly ask you to fill out this form for our project. Rest assured, your data will be kept safe and confidential with us.

danish040827@gmail.com [Switch accounts](#)

The name, email address and photo associated with your Google Account will be recorded when you upload files and submit this form.

* Indicates required question

Which category are you? *

☐ student

☐ lecture

☐ staff

☐ not all above

Figure 1.3 illustrates google form for jogging counts calories.

```

> mode_calories <- get_mode(calorie_data$calories)
> first_quartile <- quantile(calorie_data$calories, 0.25, na.rm = TRUE)
> third_quartile <- quantile(calorie_data$calories, 0.75, na.rm = TRUE)
> sd_calories <- sd(calorie_data$calories, na.rm = TRUE)
> variance_calories <- var(calorie_data$calories, na.rm = TRUE)
> # Print results
> cat("Mean Calories:", mean_calories, "\n")
Mean Calories: 318.17
> cat("Median Calories:", median_calories, "\n")
Median Calories: 289.5
> cat("Mode Calories:", mode_calories, "\n")
Mode Calories: 150
> cat("First quartile (Q1):", first_quartile, "\n")
First Quartile (Q1): 194.75
> cat("Third quartile (Q3):", third_quartile, "\n")
Third Quartile (Q3): 399
> cat("Standard Deviation:", sd_calories, "\n")
Standard Deviation: 173.2313
> cat("Variance:", variance_calories, "\n")
Variance: 30009.07

```

Figure 1.4 illustrate the mean , median , mod , first and third quartile , standard deviation and variance for calories in RStudio